Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Configuring Oracle Cloud Infrastructure (OCI) Streaming Connectors 6.1.0

This connector allows Stellar Cyber to ingest logs from the Oracle Cloud Infrastructure (OCI) and add the records to the data lake.

OCI integration with Stellar Cyber provides advanced threat detection and response capabilities, combining Oracle's security solutions with real-time threat intelligence and automated response workflows to improve case response and threat mitigation.

This connector queries the Oracle Streaming Service to collect log data found within a provided stream name. The stream can be configured for different log types.

Note the following streaming limits in the Oracle Cloud Infrastructure documentation, Limits on Streaming Resources.

This connector enables horizontal scaling of OCI connectors. It uses the consumer group feature of OCI Streaming to handle offsets and partitions. You can manually create multiple OCI connectors for a single stream with multiple partitions. It is recommended that the number of OCI connectors be less than or equal to the number of partitions. 5.5.0

Stellar Cyber connectors with the Collect function (collectors) may skip collecting some data when the ingestion volume is large, which potentially can lead to data loss. This can happen when the processing capacity of the collector is exceeded.

Connector Overview: OCI Streaming

Capabilities

- Collect: **Yes**
- Respond: **No**
- Native Alerts Mapped: **Yes**
- Runs on: **DP**
- Interval: **N/A**

Collected Data

Content Type	Index	Locating Records
Log data in provided Stream Name	Syslog	msg_class: `oracle_cloud_audit` (if `oracle.oracle.loggroupid` is _Audit)

Content Type	Index	Locating Records

		`oracle_cloud_guard` (if `oracle.source` starts with CloudGuard (case insensitive))
		`oracle_vcnflow` (for VCN flow logs)
		`oracle_cloud_waf` (if `oracle.type.keyword.is com.oraclecloud.loadbalancer.waf`) 6.1.0
		`oracle_cloud_log` (default)
		
		msg_origin.source:
		`oracle_cloud_infrastructure`
		
		msg_origin.vendor:
		`oracle`
		
		msg_origin.category:
		`paas` (default)
		`waf` (if `oracle.type.keyword.is com.oraclecloud.loadbalancer.waf`) 6.1.0

If the OCI log type is `com.oraclecloud.vcn.flowlogs.DataEvent`, VCN flow log information is reported in the Traffic Index.

Domain

N/A

Response Actions

N/A

Third Party Native Alert Integration Details

This connector ingests logs from OCI to get the raw alerts that are stored in the Syslog index.

Stellar Cyber maps OCI CloudGuard alerts. The alerts are read from the Syslog index, enriched with Stellar Cyber fields, and mapped (with deduplication) to the Alerts index.

Deduplication is by `tenantid`, `oracle.data.additionalDetails.tenantId`, `event.threat.name`, and `cloud.resource.id`.

The resource types include instances, buckets, and others.

For details, see Integration of Third Party Native Alerts.

Required Credentials and Configurations

- Bootstrap Server, Username, Auth Token, and Stream Name

Locating Records

To search the alerts in the Alerts index or to search the Original Records in the Syslog index, use the query: `msg_class: oracle_cloud_guard AND _exists_:event.threat.name`

Adding an OCI Streaming Connector

To add an OCI Streaming connector:

1. Obtain your OCI credentials

2. Add the connector in Stellar Cyber
3. Test the connector
4. Verify ingestion

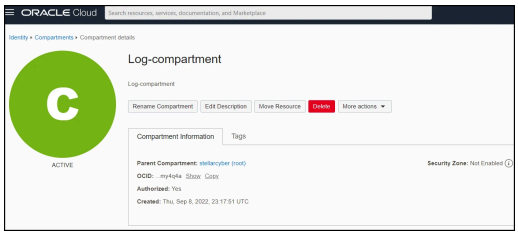
Obtaining your OCI Credentials

Follow the guidance in the Oracle documentation cited in OCI Settings to create the credentials for Stellar Cyber API calls.

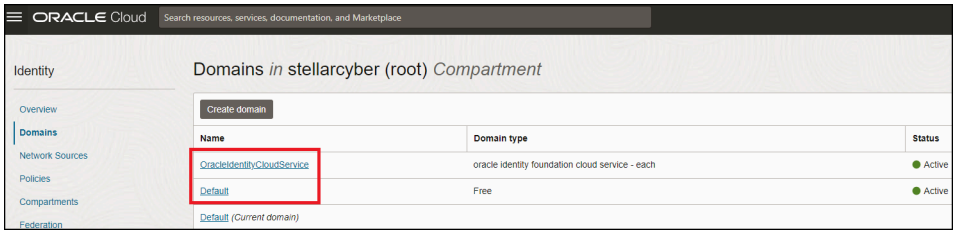
OCI Settings

To configure the OCI settings:

1. Log in as an administrative user to the Oracle Cloud console.
2. Skip this step if you have a compartment already. To create a compartment, refer to the Oracle documentation for Adding Users, search for Create a sandbox compartment, and follow the procedure. In this example, the compartment is Log-compartment.

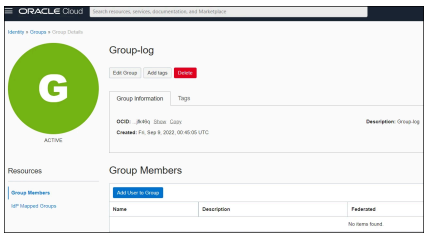


3. To create a user and group, navigate to **Domains**.

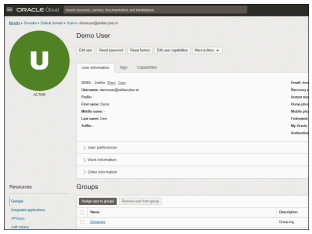


By default, there are two domains in OCI. Select the domain in which you want to create a group and user. In the next step, the example group is created in the Default domain.

4. To create a group, refer to the Oracle documentation for Adding Users, search for Create a group, and follow the procedure. In this example, the group is Group-log.



5. To create a user, refer to the Oracle documentation for Adding Users, search for Create a user, and follow the procedure. In this example, the user is demouser@stellarcyber.ai. Then, add the user to the group. In this example, the group is Group-log in the Default domain.



6. Review the Oracle documentation for How Policies Work. In the root compartment, create a policy with the following syntax:

```
Allow group <identity_domain_name>/<group_name> to <verb> <resource-type>
in compartment <compartment_name>
```

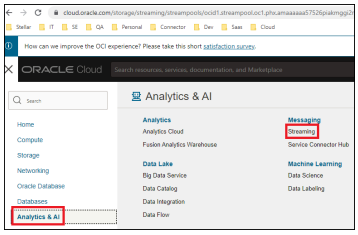
In this example, the group is Group-log in the Default domain and the compartment is Log-compartment:

```
Allow group default/Group-log to use stream-pull in compartment Log-
compartment
```

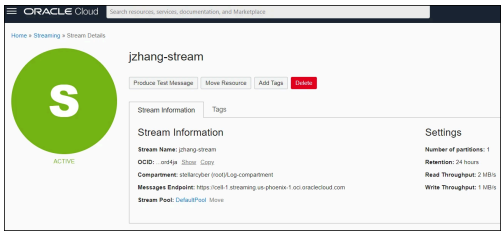
In another example, there is a user, loguser@stellarcyber, in the group Group-IDCS-log of the OracleIdentityCloudService, and the compartment is Log-compartment, so the policy will be:

```
Allow group OracleIdentityCloudService/Group-IDCS-log to use stream-pull in
compartment Log-compartment
```

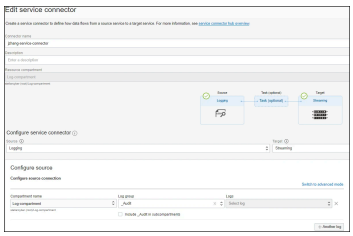
7. To set up streaming, review the Oracle documentation for Creating a Stream.
8. To create and configure the Oracle Streaming Service, navigate to **Analytics & AI** and choose **Streaming** under Messaging.



9. Click **Create Stream**. Add a Stream Name and select the compartment you created earlier. In this example, the compartment is Log-compartment. If you are creating a stream for the first time, a default Stream Pool, **DefaultPool**, is created.

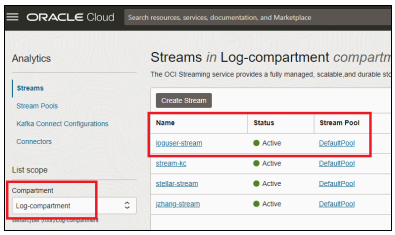


10. To create a Service Connector Hub, refer to the Oracle documentation for Creating a service connector.
11. Create the connector under the compartment you created earlier.

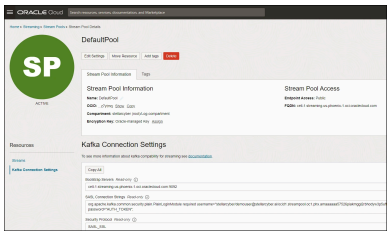


Under **Configure service connector**, note the Source, which is Logging, and the Target, which is Streaming. They will be used to configure the OCI Streaming connector with a logging source. Refer to the Oracle documentation for Creating a Connector with a Logging Source.

12. Navigate to **Streams** under Analytics and locate the stream's Stream Pool. In this example, the stream name is loguser-stream and the stream pool is the **DefaultPool**. The compartment is Log-compartment.



13. Navigate to the Kafka Connection Settings in the **DefaultPool**.



14. Configure the Kafka Connection Settings as follows:

    - The format for **Bootstrap Servers** is: `cell-1.streaming.<region>.oci.oraclecloud.com:9092`. For this example, enter: `cell-1.streaming.us-phoenix-1.oci.oraclecloud.com:9092`.

      The **Bootstrap Servers** is required for the OCI Streaming connector configuration.

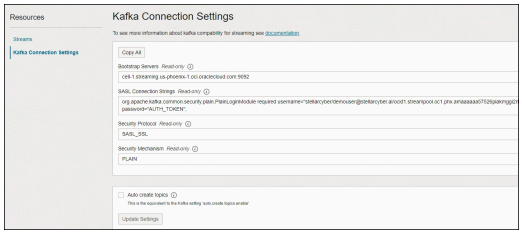
    - The format for **SASL Connection Strings** is: `username="<OCI_tenancy_name>/<domain_name>/<your_OCI_username>/<stream_pool_OCID> password="AUTH_TOKEN"`. For this example, enter:

      Username:
      `stellarcyber/Default/StreamUser/ocid1.streampool.oc1.phx.amaaaaaa57526pia6d4ng2pyndugjosjsehmibndthdxxxxxxxxxxxxxxxxx`

      or

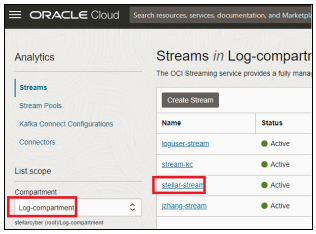
      Username:
      `stellarcyber/OracleIdentityCloudService/StreamUser/ocid1.streampool.oc1.phx.amaaaaaa57526pia6d4ng2pyndugjosjxxxxxxxxx`

      The **Username** is required for the OCI Streaming connector configuration.

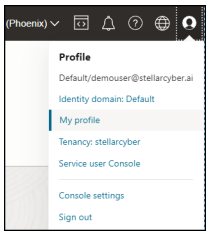


15. Navigate to **Streams** and note the stream name.

    The **Stream Name** is required for the OCI Streaming connector configuration.

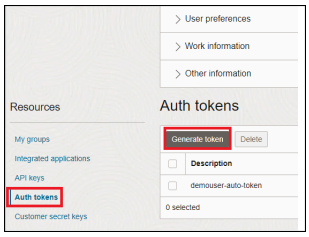


16. To obtain the authentication token, log in using the newly created user, navigate to **Profile**, and click **My profile**. In this example, the new user is demouser@stellarcyber.ai.



17. Navigate to **Auth tokens** and click **Generate token**, then note the token.

    The **Auth Token** is required for the OCI Streaming connector configuration.



Fetching Audit Logs

Once the stream is configured, you need to send audit logs to the configured stream.

Follow the guidance in the Oracle documentation for Creating a Connector with a Logging Source to transfer log data from the logging service to a target service.

You can also refer to Overview of Audit.

Log Data in the OCI Stream

Stellar Cyber ingests everything sent to the OCI stream. Most of what is ingested is stored in the Syslog index verbatim. The exceptions are:

- CloudGuard records, which Stellar Cyber normalizes. Stellar Cyber also has an alert integration with OCI CloudGuard, which is described in Integration of Third Party Native Alerts.
- VCN Flow records which Stellar Cyber normalizes and stores in the Traffic index. These records are identified by the following field:

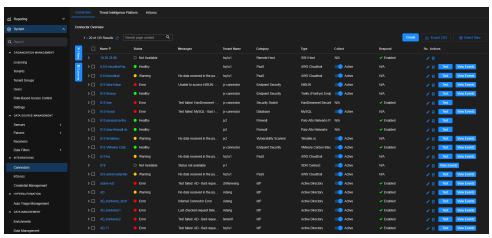
  `oracle.type: com.oraclecloud.vcn.flowlogs.DataEvent`

Refer to the Oracle documentation: Overview of Streaming or OCI Streaming FAQ.

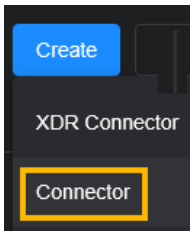
Adding the Connector in Stellar Cyber

With the access information handy, you can add an OCI Streaming connector in Stellar Cyber:

1. Log in to Stellar Cyber.
2. Click **System | INTEGRATIONS | Connectors**. The **Connector Overview** appears.

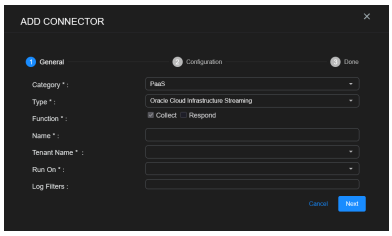


3. Click **Create** and select **Connector**.



The **General** tab of the **Add Connector** screen appears. The information on this tab cannot be changed after you add the connector.

4. Choose **PaaS** from the **Category** dropdown.
5. Choose **Oracle Cloud Infrastructure Streaming** from the **Type** dropdown.



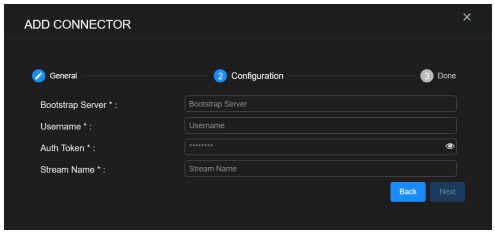
The asterisk (*) indicates a required field.

6. For this connector, the supported **Function** is **Collect**, which is enabled already.
7. Enter a **Name**.

Notes:

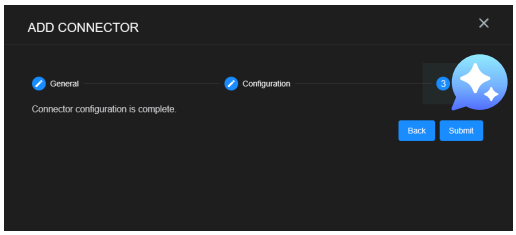
- This field does not accept multibyte characters.
- It is recommended that you follow a naming convention such as tenantname-connectortype.

8. Choose a **Tenant Name**. This identifies which tenant is allowed to use the connector.
9. Choose the device on which to run the connector.
10. (Optional) When the **Function** is **Collect**, you can apply **Log Filters**. For information, see Managing Log Filters.
11. Click **Next**. The **Configuration** tab appears.



The asterisk (*) indicates a required field.

12. Enter the **Bootstrap Server** you noted above in Obtaining your OCI Credentials.
13. Enter the **Username** you noted above.
14. Enter the **Auth Token** you noted above.
15. Enter the **Stream Name** you noted above.
16. Click **Next**. The final confirmation tab appears.



17. Click **Submit**.

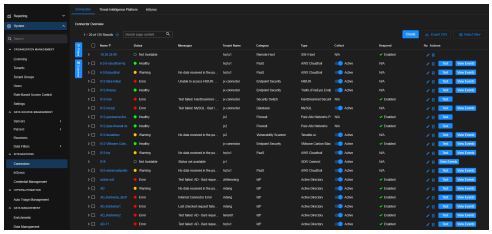
The new connector is immediately active.

Testing the Connector

When you add (or edit) a connector, we recommend that you run a test to validate the connectivity parameters you entered. The test validates authentication and connectivity.

For connectors running on a sensor, Stellar Cyber recommends that you allow 30-60 seconds for new or modified configuration details to be propagated to the sensor before performing a test.

1. Click **System | INTEGRATIONS | Connectors**. The **Connector Overview** appears.



2. Locate the connector by name that you added, or modified, or that you want to test.
3. Click **Test** at the right side of that row. The test runs immediately.

Note that you may run only one test at a time.

Stellar Cyber conducts a basic connectivity test for the connector and reports a success or failure result. A successful test indicates that you entered all of the connector information correctly.

To aid troubleshooting your connector, the dialog remains open until you explicitly close it by using the **X** button. If the test fails, you can select the button from the same row to review and correct issues.

The connector status is updated every five (5) minutes. A successful test clears the connector status, but if issues persist, the status reverts to failed after a minute.

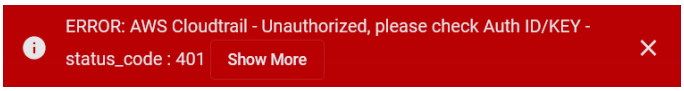
Repeat the test as needed.

Display sample messages...

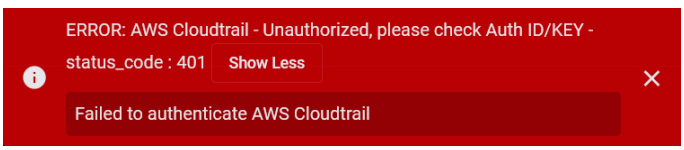
Success !



Failure with summary of issue:



Show More example detail:



If the test fails, the common HTTP status error codes are as follows:

HTTP Error Code	HTTP Standard Error Name	Explanation	Recommendation
400	Bad Request	This error occurs when there is an error in the connector configuration.	Did you configure the connector correctly?
401	Unauthorized	This error occurs when an authentication credential is invalid or when a user does not have sufficient privileges to access a specific API.	Did you enter your credentials correctly? Are your credentials expired? Are your credentials entitled or licensed for that specific resource?
403	Forbidden	This error occurs when the permission or scope is not correct in a valid credential.	Did you enter your credentials correctly? Do you have the required role or permissions for that credential?
404	Not Found	This error occurs when a URL path does not resolve to an entity.	Did you enter your API URL correctly?
429	Too Many Requests	This error occurs when the API server receives too much traffic or if a user's license or entitlement quota is exceeded.	The server or user license/quota will eventually recover. The connector will periodically retry the query. If this occurs unexpectedly or too often, work with your API provider to investigate the server limits, user licensing, or quotas.

For a full list of codes, refer to HTTP response status codes.

Verifying Ingestion

To verify ingestion:

1. Select **Threat Hunting**.

   The **Interflow Search** tab appears.

2. Change the **Indices** for the type of content you collected:
   - For all logs, change the **Indices** to **Syslog**.
   - For VCN flow logs only, change the **Indices** to **Traffic**.

   The table immediately updates to show ingested Interflow records.

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